

Project Coversheet

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Project Title	Week 2: Sales & Customer Behavior Insights – Green Cart Ltd.

Instructions:

Students must download this cover sheet, use it as the first page of their project, and then save the entire document as a PDF before submission.

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.
- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.

- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

YOU CAN START YOUR PROJECT FROM HERE

Introduction

At Green Cart Ltd., a UK-based e-commerce company specializing in ecofriendly household products, we aim to understand sales and customer behavior across regions and product lines. This analysis involves cleaning and merging sales, product, and customer datasets, creating new features, and identifying key patterns.

The insights presented will highlight revenue trends, customer purchasing behavior, product performance, and delivery outcomes. These findings will guide marketing strategies, operational decisions, and regional performance optimization for Q2.

Data Cleaning Summary

Sales Data (sales_data.csv)

- **Text columns:** delivery_status, payment_method, region standardized for consistent formatting.
- **Dates:** order_date converted to datetime.
- **Missing values:**
 - discount_applied filled with 0.
 - delivery_status and payment_method filled with “Unknown”.
 - unit_price filled with the base_price value by comparing their product_id
 - quantity filled with a minimum value 1.
 - order_id filled by generating a random 6-digit number that starts with 'O'.
- **Duplicates:** Duplicates identified via order_id; removed 2 duplicate records.
- **Numeric validation:** quantity, unit_price, and discount_applied verified as zero or greater.

Product Data (product_info.csv)

- **Text columns:** product_name, category, supplier_code standardized.
- **Dates:** launch_date converted to datetime.
- **Missing values:**
 - category filled with “Unknown”.

- **Numeric validation:** base_price verified as zero or greater.

Customer Data (customer_info.csv)

- **Text columns:** gender, region, loyalty_tier standardized.
- **Dates:** signup_date converted to datetime.
- **Missing values:**
 - loyalty_tier and gender filled with “Unknown”.
 - email missing values kept but noted.

Overall outcome:

Sales, product, and customer datasets were cleaned for analysis. Text fields and dates were standardized, missing values addressed, and duplicates removed. Numeric columns were validated to ensure non-negative values. Datasets are now consistent, complete where possible, and ready for reporting and modeling.

Feature Engineering Summary

Revenue

- 3,000 orders analyzed, total revenue £240,449.98.
- Average revenue per order £80.15, ranging from £6.80 to £249.35.
- Interquartile range: £36.53 (25%) to £116.56 (75%).

Order Week

- Extracted ISO week from order_date.
- 1,184 valid entries, ranging from week 1 to week 50.
- Median week: 23; mean week: 25.37, indicating mid-year concentration of orders.

Price Band

- Categorized unit price into Low (<£15), Medium (£15–£30), High (>£30).
- **High** being the most frequent (1,392 orders).
- Indicates a majority of purchases are higher-priced products.

Days to Order

- 1,184 valid records, mean 188 days, median 196 days.
- Range from -301 to 606 days; negative values indicate pre-launch orders.

Email Domain

- 2,947 emails analyzed across 223 unique domains.
- Most common: yahoo.com (577 customers), showing key communication channels.

Late Delivery Flag

- 3,002 records, 1,831 delivered on time.
- About 39% of orders were late, highlighting potential logistics improvements.

Overall Insights

Revenue totals £240,450.00 with most orders from high-priced products and a mid-year concentration of purchases. Days-to-order vary widely, highlighting diverse post-launch buying patterns. Yahoo.com dominates customer emails, suggesting key communication channels. About 39% of orders were delivered late, indicating room for logistics improvement.

Key Findings & Trends

1. High-Performing Products and Regions

- Cleaning products drive the most revenue (£93,787), followed by Storage and Outdoors.
- The South region generates the highest revenue (£49,774), with West and East also strong.

2. Customer Behavior and Discounts

- Discounts increase quantities sold, especially for Cleaning and Storage products.
- Gold loyalty tier customers contribute the highest revenue (£135,749), more than double silver and bronze tiers.

3. Delivery and Signup Patterns

- About 39% of orders are late, but delays are fairly balanced across regions.
- Customers signing up in peak months (July–October 2024) place more orders; November 2025 signups have the highest average order value (£100.76).

Business Question Answers

1. Which product categories drive the most revenue, and in which regions?

Cleaning products generate the highest revenue at 93,786.62, significantly outperforming all other categories, followed by Storage and Outdoors. Revenue is strongest in the South region at

49,773.50, closely followed by the West and East, while the presence of a very small Nrth value indicates a data quality issue that should be corrected.

2. Do discounts lead to more items sold?

Discounted categories generally show higher quantities sold, indicating that discounts support increased item movement rather than revenue alone. Cleaning and Storage have both the highest total discounts applied and the largest quantities sold.

3. Which loyalty tier generates the most value?

Gold loyalty customers generate the highest value by a wide margin, contributing 135,749.19 in total revenue, which is more than double the revenue of silver and bronze tiers. Silver follows with 51,571.78, while Bronze contributes 48,370.51, and the remaining tiers represent data quality issues caused by inconsistent tier labels that should be standardized before final reporting.

4. Are certain regions struggling with delivery delays?

Analysis of delivery performance across regions shows that most areas have a balanced number of on-time and late deliveries, with Central, East, North, South, and West reporting between 219 and 250 late orders each. Overall, no single region is disproportionately struggling.

5. Do customer signup patterns influence purchasing activity?

Customer signup patterns influence purchasing activity, with higher order volumes observed for signups in peak months like July – October 2024, and lower volumes in months like May 2024 and December 2025. Average order value also varies by signup month, with November 2025 showing the highest AOV (100.76), indicating that signup timing affects both order frequency and spending.

Recommendations

- Focus marketing and promotions on cleaning products and high-performing regions.
- Use discounts strategically to boost sales volume, especially for top-selling categories.
- Target Gold loyalty customers with retention campaigns and personalized offers.
- Standardize region and delivery status data to improve reporting accuracy.

- Monitor and improve logistics to reduce late deliveries, particularly during peak months.

Data Issues or Risks

One key data quality issue identified was inconsistent categorical values in fields such as region and delivery status. Variations in spelling and formatting created multiple representations of the same category, which affected aggregation and reporting accuracy.

This issue can be fixed at the source by enforcing controlled input values through dropdown selections in data entry systems and implementing automated validation checks. Regular data audits and standard reference tables would further ensure consistency over time.